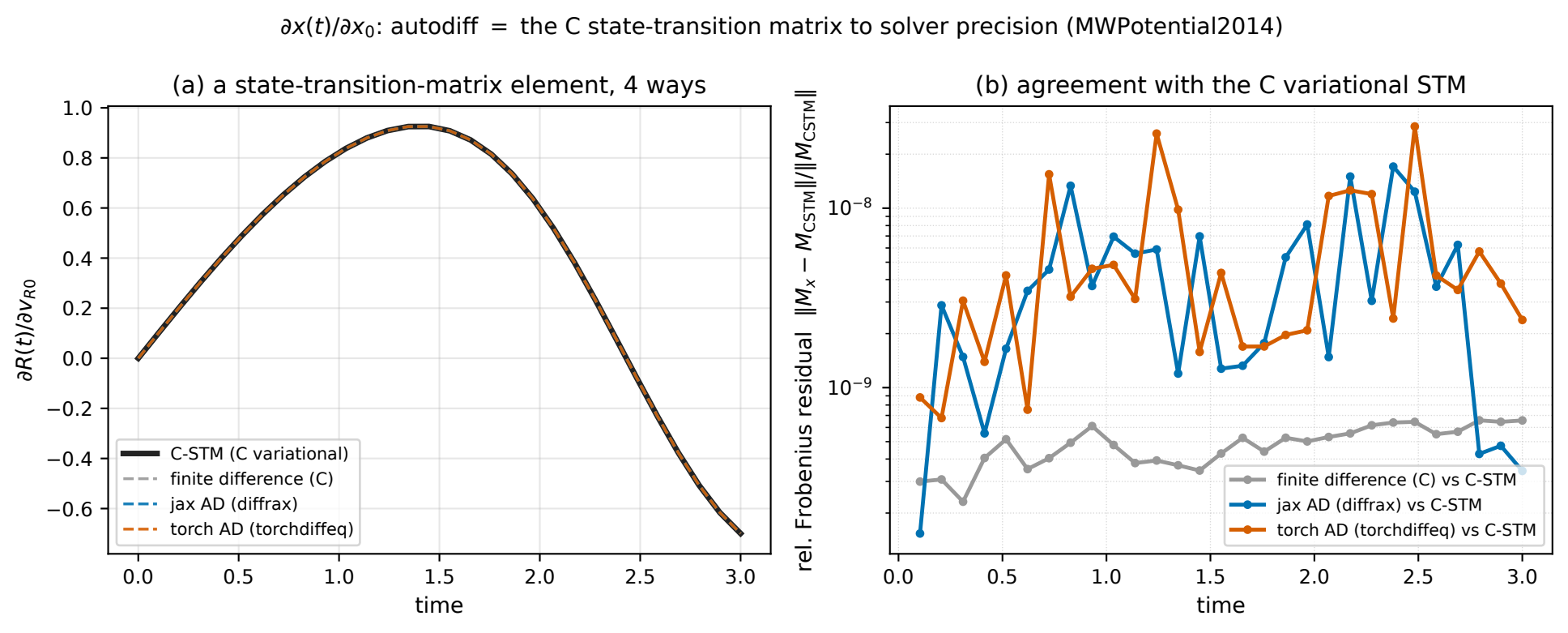
$\partial x(t)/\partial x_0$: autodiff = the C state-transition matrix to solver precision (MWPotential2014)
(a) a state-transition-matrix element, 4 ways
$\partial R(t)/\partial v_{R0}$
time
C-STM (C variational)
finite difference (C)
jax AD (diffrax)
torch AD (torchdiffeq)
(b) agreement with the C variational STM
rel. Frobenius residual $\|M_x - M_{\rm CSTM}\|/\|M_{\rm CSTM}\|$
time
finite difference (C) vs C-STM
jax AD (diffrax) vs C-STM
torch AD (torchdiffeq) vs C-STM